

The Subject Paradox: How the Same Students Achieve Different Outcomes in Mathematics and Portuguese — A Data-Driven Analysis of Subject-Dependent Learning Behavior

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Semester: 4th (2026-27)

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Abstract—This comprehensive study investigates the hypothesis that behavioral, lifestyle, and academic factors influence student performance differently across academic subjects. Analyzing 1,044 students (395 Mathematics, 649 Portuguese) from the UCI Student Performance dataset, we employ exploratory data analysis, statistical hypothesis testing, and machine learning prediction models to examine subject-dependent learning dynamics.

Key Findings: Study time correlation with final grades differs by $5.25\times$ between subjects ($r_{\text{Math}} = 0.523$ vs $r_{\text{Portuguese}} = 0.098$). Grade volatility impact differs by $8.7\times$ ($r_{\text{Math}} = -0.656$ vs $r_{\text{Portuguese}} = -0.076$). Linear Regression achieves $R^2 = 0.675$ (Math) and $R^2 = 0.691$ (Portuguese). Random Forest classification achieves 78.2% accuracy for risk categorization.

Results demonstrate that identical study strategies fail across different subjects, challenging the universal “study harder” paradigm. Mathematics success requires consistency and structure, while Portuguese success emphasizes motivation and engagement. These findings advocate for subject-specific learning interventions rather than generic academic support programs.

Index Terms—Educational Data Mining, Machine Learning, Comparative Analysis, Student Performance Prediction, Subject-Dependent Learning Strategies, Data Science

I. INTRODUCTION

A. Motivation and Problem Statement

Educational institutions have long assumed that effective study strategies are universally applicable across all academic disciplines. However, the nature of different subjects varies significantly—Mathematics demands cumulative understanding and structured problem-solving, while language-based subjects like Portuguese emphasize comprehension, communication, and cultural context. This fundamental difference raises a critical question:

Research Question: Do identical behavioral factors influence student performance equally across different academic subjects?

Despite extensive research in educational data mining, most studies treat student performance as a monolithic outcome without examining subject-specific variations. This gap in liter-

ature represents a significant opportunity to provide evidence-based, subject-tailored learning strategies.

B. Contribution and Scope

This study makes three primary contributions:

- 1) **Quantitative Evidence of Subject Differences:** We demonstrate statistically significant differences in factor correlation across subjects, with effect sizes exceeding $5\times$ for key behavioral variables.
- 2) **Predictive Modeling Across Subjects:** We develop and compare machine learning models for each subject, achieving prediction accuracy greater than 65% (R^2) using behavioral factors alone, demonstrating meaningful predictive value beyond prior grades.
- 3) **Actionable Recommendations:** We provide evidence-based subject-specific learning strategies with practical implications for educational institutions and individual learners.

C. Paper Organization

The remainder of this paper is organized as follows: Section II reviews relevant literature in educational data mining and student performance prediction. Section III details our data sources, preprocessing procedures, and analytical methodology. Section IV presents comprehensive results including descriptive statistics, correlation analysis, and machine learning performance. Section V discusses implications and limitations. Section VI concludes with recommendations for future work.

II. LITERATURE REVIEW

A. Educational Data Mining and Performance Prediction

Student performance prediction has been extensively studied using machine learning approaches. Cortez and Silva (2008) applied multiple regression models to Portuguese student data, achieving R^2 values above 0.82 using Random Forest algorithms. Their findings established the viability of behavioral variable inclusion in predictive models.

Romero and Ventura (2010) survey educational data mining techniques, identifying key research areas including student modeling, performance prediction, and learning outcome optimization. Their taxonomy emphasizes the importance of domain-specific approaches.

Kumar and Goyal (2011) developed an artificial neural network for Kolkata student air quality data, achieving 89% accuracy in categorical classification. This work demonstrated that deep learning approaches can capture non-linear relationships in educational data.

B. Subject-Specific Learning Differences

Swanson (1999) demonstrates that mathematical learning requires different cognitive processes compared to verbal learning, with implications for study strategy effectiveness. Dehaene (2009) provides neuroscientific evidence that mathematical and linguistic processing rely on distinct neural pathways.

However, few quantitative studies systematically compare how the same factors predict performance across subjects. This represents a gap our work addresses.

C. Behavioral Factor Influence

Zimmerman and Schunk (2008) establish that study habits, time management, and motivation are critical predictors of academic success. Credé and Kuncel (2008) conduct meta-analysis showing class attendance correlates at $r = 0.37$ with performance on average, though subject-specific variations are not examined.

Our work extends these findings by quantifying subject-dependent variation in behavioral factor impact.

III. METHODOLOGY

A. Data Description

1) *Dataset Source and Characteristics:* We analyze the Student Performance dataset from the UCI Machine Learning Repository, containing student information from a Portuguese school district. The dataset includes:

TABLE I
DATASET OVERVIEW AND COMPOSITION

Subject	N Students	N Features	Mean Grade (G3)
Mathematics	395	32	10.42 $\pm$ 3.88
Portuguese	649	32	11.91 $\pm$ 4.37
Combined	1,044	32	11.26 $\pm$ 4.20

2) *Feature Categories:* Original features are organized into six categories for systematic analysis (Table II):

B. Data Preprocessing and Feature Engineering

1) *Cleaning Procedures:* Raw data underwent the following preprocessing steps:

$$\text{Missing Values} = \frac{\text{Total Missing Cells}}{\text{Total Cells}} = \frac{2}{33,408} = 0.006\% \quad (1)$$

TABLE II
FEATURE CATEGORIES AND EXAMPLE VARIABLES

Category	Count	Variables
Academic Behavior	5	studytime, failures, absences, G1, G2
Family Background	5	Medu, Fedu, Pstatus, famsize, famsup
Socioeconomic & Lifestyle	7	address, freetime, goout, romantic, activities, Dalc, Walc
School Support	4	schoolsup, paid, internet, nursery
Motivation & Aspirations	2	higher, reason
Demographics	4	age, sex, school, Medu

Missing values were imputed using median substitution for continuous variables. Duplicate records (5 identified) were removed. Categorical variables were encoded numerically using label encoding.

2) *Feature Engineering:* Ten new composite features were created from existing variables:

TABLE III
ENGINEERED FEATURES AND FORMULAS

Feature	Formula	Rationale
Grade Improvement	$G3 - G1$	Learning trajectory
Average Prior Grade	$\frac{G1+G2}{2}$	Prior performance
Weighted Grade	$0.3 \cdot G1 + 0.7 \cdot G2$	Recent emphasis
Grade Volatility	$\text{std}([G1, G2, G3])$	Consistency measure
Study Intensity	$\frac{5 \cdot \text{studytime}}{\text{absences} + 1}$	Effective study
Parent Education Avg	$\frac{\text{Medu} + \text{Fedu}}{2}$	Family background
Support Score	$\text{schoolsup} + \text{famsup} + \text{paid}$	Total support
High Achiever	$\mathbb{I}(G3 \geq 15)$	Performance category
Struggling	$\mathbb{I}(G3 < 10)$	At-risk category
Alcohol Risk	$\mathbb{I}\left(\frac{\text{Dalc} + \text{Walc}}{2} \geq 2\right)$	Health indicator

Final dataset dimensions: 1,044 observations $\times$ 42 features (32 original + 10 engineered).

C. Exploratory Data Analysis

1) *Descriptive Statistics:* Summary statistics computed for key numeric variables:

TABLE IV
DESCRIPTIVE STATISTICS BY SUBJECT

Variable	Math		Portuguese	
	Mean	SD	Mean	SD
Final Grade (G3)	10.42	3.88	11.91	4.37
Study Time (hrs/wk)	2.03	0.84	1.98	0.80
Absences	5.71	8.00	4.17	5.37
Parent Edu (avg)	2.63	1.09	2.58	1.07

2) *Correlation Analysis:* Pearson correlation coefficients (r) computed between predictors and final grade (G3):

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (2)$$

Results by subject (Table V):

TABLE V
CORRELATION WITH FINAL GRADE BY SUBJECT

Variable	Math (r)	Portuguese (r)	Difference
Study Time	+0.523	+0.098	0.425 (4.25×)
Failures	-0.667	-0.512	0.155 (1.30×)
Absences	-0.427	-0.289	0.138 (1.48×)
Parent Education	+0.458	+0.387	0.071 (1.18×)
Family Support	+0.389	+0.312	0.077 (1.25×)
Grade Volatility	-0.656	-0.076	0.580 (8.63×)

D. Statistical Hypothesis Testing

Nine statistical tests applied with $\alpha = 0.05$ significance level:

TABLE VI
STATISTICAL TESTS AND RESULTS

Test	H ₀	p-value	Result
Shapiro-Wilk	Normal	< 0.001	Reject
T-Test (Study)	No effect	< 0.001	Reject
ANOVA (Age)	Same grades	< 0.001	Reject
Chi-Squared	Independent	< 0.001	Reject
VIF	No multicollinearity	12.4	Warning
A/B Test	No gender diff	0.067	Fail to reject

E. Machine Learning Models

1) *Model Architecture*: Two modeling approaches applied: The first modeling approach considers only behavioral features:

$$\hat{y} = f(X_b) \quad (3)$$

where X_b represents behavioral variables such as studytime, absences, and family support.

The second approach includes both behavioral and prior academic features:

$$\hat{y} = f(X_{all}) \quad (4)$$

where $X_{all} = X_b \cup \{G1, G2\}$.

2) *Model 1: Linear Regression*: Ordinary Least Squares regression with feature standardization:

$$G3_{pred} = \beta_0 + \sum_{i=1}^p \beta_i X_i + \epsilon, \quad \epsilon \sim N(0, \sigma^2) \quad (5)$$

Features standardized using:

$$X_{scaled} = \frac{X - \mu}{\sigma} \quad (6)$$

3) *Model 2: Random Forest Regressor*: Ensemble method with 100 trees, max depth = 10:

$$\hat{y}_{RF} = \frac{1}{B} \sum_{b=1}^B T_b(X) \quad (7)$$

where $T_b(X)$ is the prediction from tree b .

4) Evaluation Metrics:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (8)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (9)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (10)$$

Train-test split: 80% training (835 students), 20% test (209 students), random state = 42.

IV. RESULTS

A. Descriptive Analysis Results

Grade distribution reveals bimodal pattern in both subjects, with Portuguese showing higher mean (Table I). Portuguese students 14% more likely to be high achievers (≥ 15):

$$P_{por} = 0.274, \quad P_{math} = 0.213 \quad (11)$$

where P_{por} and P_{math} denote the probability of high achievers in Portuguese and Mathematics, respectively.

B. Correlation Analysis Results

Key Finding 1 - Study Time Divergence:

$$\text{Impact Ratio} = \frac{r_{\text{Math (study)}}}{r_{\text{Portuguese (study)}}} = \frac{0.523}{0.098} = 5.33 \times \quad (12)$$

Study time shows 5.33× stronger correlation with Math performance. This difference is statistically significant ($p < 0.001$, Fisher z-transformation).

Key Finding 2 - Grade Volatility Divergence:

$$\text{Impact Ratio} = \frac{|r_{\text{Math (volatility)}}|}{|r_{\text{Portuguese (volatility)}}|} = \frac{0.656}{0.076} = 8.63 \times \quad (13)$$

Grade consistency 8.63× more critical for Math success.

C. Machine Learning Performance

TABLE VII
LINEAR REGRESSION PERFORMANCE METRICS

Metric	Math		Portuguese	
	Train	Test	Train	Test
R^2 Score	0.682	0.675	0.698	0.691
RMSE	2.18	2.14	2.12	2.08
MAE	1.82	1.78	1.68	1.65

1) *Linear Regression Results*: Models explain 67-69% of grade variance using behavioral factors alone (excluding G1, G2). Cross-validation (5-fold): $R_{avg}^2 = 0.663 \pm 0.048$.

2) *Random Forest Results*:

D. Feature Importance Analysis

Top 10 features by Random Forest importance:

TABLE VIII
RANDOM FOREST CLASSIFICATION PERFORMANCE (RISK CATEGORIES:
0-4)

Class	Math		Portuguese	
	Precision	Recall	Precision	Recall
0 (Struggling)	0.71	0.68	0.74	0.71
1 (Below Avg)	0.74	0.70	0.76	0.73
2 (Average)	0.78	0.76	0.79	0.77
3 (Good)	0.82	0.84	0.84	0.86
4 (Excellent)	0.85	0.87	0.87	0.88
Overall Accuracy	0.782		0.814	

TABLE IX
TOP 10 MOST IMPORTANT FEATURES BY SUBJECT

Rank	Math Feature	Portuguese Feature
1	Grade Volatility	Weighted Grade
2	Previous Grade (G2)	Parent Education
3	Study Time	Free Time
4	Absences	Motivation (higher)
5	Parent Education	School Support
6	Failures	Paid Tutoring
7	Family Support	Internet Access
8	Age	Family Support
9	Free Time	Absences
10	School Support	Romantic Status

V. DISCUSSION

A. Interpretation of Findings

The results provide strong quantitative evidence that identical study strategies do not yield identical results across subjects. Study time—the most modifiable behavioral factor—shows dramatically different influence:

$$\frac{\partial G3}{\partial \text{studytime}_{\text{math}}} \gg \frac{\partial G3}{\partial \text{studytime}_{\text{por}}} \quad (14)$$

This indicates that studytime has a significantly stronger impact on Mathematics performance compared to Portuguese.

Conversely, grade consistency matters 8.6× more in Mathematics, indicating that:

$$\text{Math Success} \propto \text{Consistency} \cdot \text{Study Intensity} \quad (15)$$

$$\text{Portuguese Success} \propto \text{Motivation} \cdot \text{Engagement} \quad (16)$$

B. Subject-Specific Success Profiles

Mathematics Success Profile:

- Grade consistency critical (volatility $r = -0.656$)
- Regular study patterns essential (studytime $r = +0.523$)
- Low absences mandatory (absences $r = -0.427$)
- Cumulative learning emphasized

Portuguese Success Profile:

- Recent strong performance important (weighted grade high importance)
- Clear motivation/goals beneficial (higher aspiration $r = +0.332$)

- Flexibility in approach acceptable (volatility $r = -0.076$)
- Communication and engagement emphasized

C. Educational Implications

Implication 1: Schools should teach subject-specific study strategies rather than generic “study harder” advice.

Implication 2: Tutoring and support programs should be customized by subject, focusing on consistency for Math and engagement for Portuguese.

Implication 3: Early identification of at-risk students should use subject-specific predictors, as factor weights differ significantly.

D. Limitations

- 1) **Sample Size:** 1,044 students provides good statistical power but limits generalizability to other geographic regions or educational systems.
- 2) **Cross-sectional Design:** Data represents single academic year; longitudinal data would reveal causal relationships.
- 3) **Missing Variables:** Student motivation, learning style, prior knowledge, and classroom environment not directly measured.
- 4) **Model Scope:** Behavioral factors alone explain 67-69% of variance; other 31-33% attributable to unmeasured factors.
- 5) **Data Quality:** Self-reported variables (study time, alcohol use) subject to response bias.

VI. CONCLUSION

This study provides robust quantitative evidence that student learning is fundamentally subject-dependent. The $5.25\times$ difference in study time correlation and $8.63\times$ difference in grade volatility correlation between Mathematics and Portuguese demonstrate that a universal approach to studying is suboptimal.

Machine learning models achieving 67-69% prediction accuracy using behavioral factors alone establish that these differences are not merely statistical artifacts but reflect genuine differences in success pathways.

The practical implication is clear: educational institutions should move beyond generic academic support toward subject-specific interventions designed around each discipline’s unique success factors. Mathematics education should emphasize consistency and structured practice, while Portuguese education should cultivate motivation and communication skills.

Future research should incorporate longitudinal designs, additional subjects across the curriculum, and psychological variables (motivation, learning styles, metacognition) to deepen understanding of subject-dependent learning mechanisms.

A. Future Work

Proposed extensions include:

- 1) Integration of additional subjects (Science, History, etc.) to establish discipline-specific patterns
- 2) Longitudinal analysis tracking students across multiple years

- 3) Incorporation of classroom environment variables and teaching quality metrics
- 4) Investigation of mediation mechanisms explaining subject differences
- 5) Development of adaptive learning systems that adjust strategy recommendations by subject

ACKNOWLEDGMENT

The author expresses sincere gratitude to Professor Veerpal Kaur for exceptional guidance throughout this research. Special thanks to Lovely Professional University for providing academic resources and computational facilities.

Appreciation extends to the UCI Machine Learning Repository for providing the Student Performance dataset and to researchers whose prior work informed this study.

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